

CS5275 Problem Set 3

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Problem 1

a. WLOG assume that $\text{vol}(S) \leq \text{vol}(V \setminus S)$, so that

$$\Phi(S) = \frac{|E(S, V \setminus S)|}{\text{vol}(S)} = \phi.$$

Consider a probability distribution $\mathbf{x}$ supported on S , defined by

$$\mathbf{x}(v) = \begin{cases} \frac{\deg(v)}{\text{vol}(S)}, & v \in S, \\ 0, & v \notin S. \end{cases}$$

Notice that $\mathbf{x}(A) = \frac{\text{vol}(A)}{\text{vol}(S)}$ for $A \subseteq S$.

For a standard (non-lazy) random walk on G , $(\mathbf{x}_0, \mathbf{x}_1, \dots)$, define a ‘killed mass’ process $\mathbf{m}_i(v)$ for $i \geq 0$ and $v \in S$, as suggested in the hint, by

$$\mathbf{m}_i(v) := \Pr[\mathbf{x}_i = v \wedge \mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{i-1} \in S],$$

i.e., $\mathbf{m}_i$ is the (unnormalized) distribution of the walk at time i conditioned on having stayed inside S up to time i , or the amount of ‘surviving’ probability mass at time i in the ‘killed’ walk. Then we have the following lemma.

Lemma 1. *For all $i \geq 0$ and $v \in S$, we have $\mathbf{m}_i(v) \leq \frac{\deg(v)}{\text{vol}(S)}$.*

Proof. We prove by induction. The base case $i = 0$ is trivial as $\mathbf{m}_0(v) = \mathbf{x}(v)$.

For the inductive step, assume that $\mathbf{m}_i(v) \leq \frac{\deg(v)}{\text{vol}(S)}$. By definition

$$\begin{aligned} \mathbf{m}_{i+1}(v) &= \sum_{u \in S} \mathbf{m}_i(u) \Pr[u \rightarrow v] \\ &= \sum_{u \in S, \{u, v\} \in E} \mathbf{m}_i(u) \frac{1}{\deg(u)} \\ &\leq \sum_{u \in S, \{u, v\} \in E} \frac{\deg(u)}{\text{vol}(S)} \frac{1}{\deg(u)} \\ &= \frac{1}{\text{vol}(S)} \sum_{u \in S, \{u, v\} \in E} 1 \leq \frac{\deg(v)}{\text{vol}(S)}. \end{aligned}$$

Thus the claim holds. □

Now we prove that each step ‘leaks’ at most ϕ mass from the killed mass process. For each $v \in S$, consider the number of edges from v that go outside S , denoted as $\text{out}(v) := |\{u \notin S : \{u, v\} \in E\}|$. Then $\sum_{v \in S} \text{out}(v) = |E(S, V \setminus S)|$, and the probability that the random walk exits S at time $i + 1$ is $p_{i+1} = \sum_{v \in S} \mathbf{m}_i(v) \frac{\text{out}(v)}{\deg(v)}$.

Via Lemma 1, we have

$$p_{i+1} \leq \sum_{v \in S} \frac{\deg(v)}{\text{vol}(S)} \frac{\text{out}(v)}{\deg(v)} = \frac{\sum_{v \in S} \text{out}(v)}{\text{vol}(S)} = \frac{|E(S, V \setminus S)|}{\text{vol}(S)} = \phi.$$

Therefore, starting from $\mathbf{x}$, the probability of exiting by time t is at most

$$\Pr[\text{exit by time } t] = \sum_{i=0}^{t-1} p_{i+1} \leq \sum_{i=0}^{t-1} \phi = t\phi \leq 0.1.$$

Recall that we are asked to show that

$$\Pr[\text{a } t\text{-step random walk started at } v \text{ never leaves } S] \geq \frac{1}{2}.$$

Denote the probability above starting from some $v \in S$ as $p(v)$, and $q(v) := 1 - p(v)$. Then we have $\mathbb{E}_{v \sim \mathbf{x}}[q(v)] \leq 0.1$.

Define the set of ‘bad’ vertices $Q := \{v \in S : q(v) > 1/2\}$. Since every $v \in Q$ contributes more than $1/2$ to the expectation,

$$\mathbb{E}_{v \sim \mathbf{x}}[q(v)] = \sum_{v \in S} \mathbf{x}(v) q(v) \geq \sum_{v \in Q} \mathbf{x}(v) q(v) > \frac{1}{2} \sum_{v \in Q} \mathbf{x}(v) = \frac{1}{2} \mathbf{x}(Q) \implies \mathbf{x}(Q) < 0.2.$$

But then

$$\mathbf{x}(Q) = \frac{\text{vol}(Q)}{\text{vol}(S)} < 0.2 \implies \text{vol}(Q) < 0.2 \text{vol}(S),$$

so taking $X = S \setminus Q = \{v \in S : q(v) \leq 1/2\}$, we have $\text{vol}(X) = \text{vol}(S) - \text{vol}(Q) \geq 0.8 \text{vol}(S)$, and for every $v \in X$,

$$\Pr[\text{a } t\text{-step random walk started at } v \text{ never leaves } S] = 1 - q(v) \geq \frac{1}{2},$$

as required.

Problem 2

- a. Denote the adjacency matrix of G as $\mathbf{A}$, with eigenvalues $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq -d$. Recall that Ramanujan graphs satisfy

$$\sigma_2 = \max\{|\lambda_2|, |\lambda_3|, \dots, |\lambda_n|\} \leq 2\sqrt{d-1}.$$

For any cut $(S, V \setminus S)$, denote $s := |S|$ and $\mathbf{x}$ to be the indicator vector for S . Decompose $\mathbf{x}$ into two orthogonal components: one parallel to the all-ones vector $\mathbf{1}$, and one perpendicular to it. Thus

$$\mathbf{x} = \frac{s}{n}\mathbf{1} + \mathbf{v}.$$

By definition, $\mathbf{v} \perp \mathbf{1}$ and

$$\|\mathbf{v}\|^2 = \|\mathbf{x}\|^2 - \left\|\frac{s}{n}\mathbf{1}\right\|^2 = s - n\left(\frac{s}{n}\right)^2 = s - \frac{s^2}{n} = \frac{s(n-s)}{n}.$$

Note that $\mathbf{x}^\top \mathbf{A} \mathbf{x}$ is just $e(S, S)$. Substituting gives

$$\begin{aligned} \mathbf{x}^\top \mathbf{A} \mathbf{x} &= \left(\frac{s}{n}\mathbf{1} + \mathbf{v}\right)^\top \mathbf{A} \left(\frac{s}{n}\mathbf{1} + \mathbf{v}\right) \\ &= \left(\frac{s}{n}\right)^2 \mathbf{1}^\top \mathbf{A} \mathbf{1} + \mathbf{v}^\top \mathbf{A} \mathbf{v} \\ &= \frac{s^2}{n^2}(dn) + \mathbf{v}^\top \mathbf{A} \mathbf{v} = \frac{ds^2}{n} + \mathbf{v}^\top \mathbf{A} \mathbf{v}. \end{aligned}$$

But as $\mathbf{v} \perp \mathbf{1}$, the term $\mathbf{v}^\top \mathbf{A} \mathbf{v}$ is lower bounded in terms of σ_2 as

$$\mathbf{v}^\top \mathbf{A} \mathbf{v} \geq -\sigma_2 \|\mathbf{v}\|^2 = -\sigma_2 \frac{s(n-s)}{n}.$$

Thus we have

$$e(S, S) \geq \frac{ds^2 - \sigma_2 s(n-s)}{n}.$$

Since G is d -regular, we have $\text{vol}(S) = ds$. But $\text{vol}(S)$ is just the sum of the degrees of all the vertices in S , which include (i) all edges crossing the cut $(S, V \setminus S)$ and (ii) all internal edges in S . Thus

$$ds = |E(S, V \setminus S)| + e(S, S).$$

Rearranging gives

$$\begin{aligned} |E(S, V \setminus S)| &= ds - e(S, S) \\ &\leq ds - \left(\frac{ds^2 - \sigma_2 s(n-s)}{n}\right) \\ &= (d + \sigma_2) \frac{s(n-s)}{n}. \end{aligned}$$

Notice that $s(n-s)/n$ has maximum value $n/4$ attained at $s = n/2$. Together with $\sigma_2 \leq 2\sqrt{d-1}$, we conclude that

$$\begin{aligned} |E(S, V \setminus S)| &\leq \left(d + 2\sqrt{d-1}\right) \frac{n}{4} \\ &= \frac{nd}{4} + \frac{1}{2}n\sqrt{d-1} \\ &\in \frac{nd}{4} + O\left(n\sqrt{d}\right), \end{aligned}$$

as required.

b. *Actually, this part holds for any d -regular graph, not just Ramanujan.*

Recall that a d -regular graph $G = (V, E)$ has $|E| = nd/2$.

Construct a cut $(S, V \setminus S)$ as follows: for every $v \in V$, put v into S with probability $1/2$ and into $V \setminus S$ with probability $1/2$.

For every $e = \{u, v\} \in E$, consider the event ' $u \in S, v \in V \setminus S$ or $v \in S, u \in V \setminus S$ '. Denote its indicator random variable as $\mathbb{1}_e$. Then $\mathbb{E}[\mathbb{1}_e] = 1/2$ by definition.

Therefore

$$\mathbb{E}[|E(S, V \setminus S)|] = \sum_{e \in E} \mathbb{E}[\mathbb{1}_e] = \frac{1}{2}|E| = \frac{nd}{4},$$

so there must exist some S such that

$$|E(S, V \setminus S)| \geq \frac{nd}{4} \in \frac{nd}{4} - O(n\sqrt{d}),$$

as required.

Problem 3

- a. Recall that the vertex connectivity $\kappa(G)$ is the minimum size of a vertex cut. Take $C \subseteq V$ to be a minimum vertex cut, so that $\kappa(G) = |C|$ and $G - C$ is disconnected. Let L be its smallest connected component, so that $0 < |L| \leq |V \setminus C|/2 \leq |V|/2$ and there are no edges between L and $(V \setminus C) \setminus L$.

Given that

$$\Phi(G) = \min_{S \subseteq V: 0 < |S| \leq |V|/2} \frac{|E(S, V \setminus S)|}{\text{vol}(S)} \geq \phi,$$

substituting $S = L$ gives $|E(L, V \setminus L)| \geq \phi \text{vol}(L)$.

By definition, $|E(L, V \setminus L)| = |E(L, C)|$. We also have $|E(L, C)| \leq |L||C|$ as G is simple and $\text{vol}(L) = d|L|$ as G is d -regular.

Therefore, $|L||C| \geq \phi d|L|$, so that $\kappa(G) = |C| \geq \phi d \in \Omega(\phi d)$ as required.

Problem 4

- a. Consider the following algorithm.

Algorithm 1 Finding a minimum vertex cut with a promise

Require: A graph $G = (V, E)$.

Some $S \subseteq V$ such that $|S| = k + 1$ where $k = \lfloor \sqrt{n} \rfloor$ and $n = |V|$.

A graph $G' = (V \cup \{r\}, E \cup \{\{s, r\} : s \in S\})$.

An oracle $\text{Oracle}(H, s, t)$ that computes an s - t minimum vertex cut for $H = G$ or $H = G'$.

Ensure: Finds a minimum vertex cut of G or correctly reports that $\kappa(G) > k$, using $O(n)$ oracle queries and $O(n^2)$ extra time.

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 $C_{\min} \leftarrow V$   $\triangleright$  the current minimum vertex cut of  $G$ 
for all  $u, v \in S$  with  $u \neq v$ ,  $\{u, v\} \notin E[G[S]]$  do
     $\triangleright$  loop 1: iterate over all possible unordered pairs of vertices in  $S$ 
     $C \leftarrow \text{Oracle}(G, u, v)$   $\triangleright$  query inside  $G$ 
    if  $|C| < |C_{\min}|$  then
         $C_{\min} \leftarrow C$   $\triangleright$  update when a smaller one is found
    end if
end for
for all  $v \in V \setminus S$  do  $\triangleright$  loop 2: iterate over all vertices in  $V \setminus S$  with  $r$ 
     $C \leftarrow \text{Oracle}(G', r, v)$   $\triangleright$  query inside  $G'$ 
    if  $|C| < |C_{\min}|$  then
         $C_{\min} \leftarrow C$   $\triangleright$  update when a smaller one is found
    end if
end for
if  $|C_{\min}| \leq k$  then  $\triangleright$  report a minimum vertex cut
    return  $C_{\min}$ 
else  $\triangleright$  report otherwise
    return ' $\kappa(G) > k$ '
end if

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It is easy to see that, even in the worst case, the first loop takes at most $O(k^2) = O(n)$ oracle queries on G and the second loop takes $n - (k + 1) \in O(n)$ oracle queries on G' . Construction of G' and looping over all pairs/vertices takes $O(n^2)$ extra time in the worst case. Thus the time complexity constraint is satisfied.

We now show that the algorithm always gives a correct output. First, we prove a useful lemma.

Lemma 2. *In the algorithm, all vertex cuts from oracle calls have size $\geq \kappa(G)$.*

Proof. In the first loop, any oracle call on (G, u, v) returns a set C whose size is $\kappa_G(u, v)$ by definition. Thus $|C| = \kappa_G(u, v) \geq \min_{s \neq t} \kappa_G(s, t) = \kappa(G)$.

In the second loop, any oracle call on (G', r, v) returns a set C with $|C| \leq k$. As $|S| = k + 1$, there exists some s' in S but not in C . We show that C is also a vertex cut in G , so that $|C| \leq \kappa(G)$. Suppose otherwise, so that $G - C$ is connected and there exists a path from s' to v . From the construction of G' , r is adjacent to s' , so in $G' - C$ there exists a path from r to v , which gives a contradiction. $\square$

Now let C be a minimum vertex cut of G such that $|C| \leq k$. By definition, removing C from G partitions the remaining vertices into at least two non-empty, disconnected components. Denote one of these components by L and the union of all other components by R . Consider the two possible cases:

- (a) S intersects at least two disconnected components of $G \setminus C$.

For $u, v \in S$, WLOG say that $u \in L$ and $v \in R$. Because u and v are in disconnected components of $G \setminus C$, any path between u and v in G must pass through C . Thus, C is a u - v vertex cut in G , which will be found during the first loop in our algorithm.

- (b) S intersects at most one disconnected component of $G \setminus C$.

WLOG say that S intersects L but not R . Thus $S \subseteq L \cup C$, so that any $v \in R$ also satisfies $v \in V \setminus S$. In G' , because r is connected only to the vertices in S , any path from r to v must pass through S , and consequently through $L \cup C$. But since there are no direct edges between L and R in G , any path originating from L to $v \in R$ must pass through C . Therefore, any path from r to v in G' must also pass through C . Thus, C is an r - v vertex cut in G' , which will be found during the second loop in our algorithm.

Thus in both cases, if such a C exists, the algorithm is guaranteed to find it. By Lemma 2, this covers the case where $\kappa(G) \leq k$. Otherwise we must have $\kappa(G) > k$, which is what the algorithm reports if $|C_{\min}| > k$ at the end.

Thus the algorithm either finds a minimum vertex cut of G or correctly reports that $\kappa(G) > k$, as required.